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Studierichting: Informatica
Oefeningenreeks 1C nr. 55

Opgave: $1 + \operatorname{cis} x \cdot \operatorname{cis} 2x \cdot \operatorname{cis} 3x = 0$

$\Rightarrow 1 + (\cos x + i \sin x)(\cos 2x + i \sin 2x)(\cos 3x + i \sin 3x) = 0$

$\Rightarrow 1 + (\cos x \cdot \cos 2x + i \sin 3x - \sin x \cdot \sin 2x)(\cos 3x + i \sin 3x) = 0$

$\Rightarrow 1 + (\cos 3x + i \sin 3x)(\cos 3x + i \sin 3x) = 0$

$\Rightarrow 1 + \cos^2 3x + 2i \cos 3x \cdot \sin 3x - \sin^2 3x = 0$

$\Rightarrow 1 + \cos^2 3x + i \sin 6x - \sin^2 3x = 0$

$\Rightarrow 1 + \cos 6x + i \sin 6x = 0$

$a = 1 + \cos 6x,\ b = \sin 6x$

$r = \sqrt{a^2 + b^2},\ \tan\theta = \dfrac{b}{a} \Rightarrow \theta = \operatorname{Bgtan} \dfrac{b}{a}$

$\Rightarrow r = \sqrt{(1 + \cos 6x)^2 + \sin^2 6x} = \sqrt{1 + 2\cos 6x + \cos^2 6x + \sin^2 6x}$

$= \sqrt{2 + 2\cos 6x} = \sqrt{2} \cdot \sqrt{1 + \cos 6x}$

$\tan\theta = \dfrac{\sin 6x}{1 + \cos 6x}$

$= \dfrac{2 \sin 3x \cos 3x}{(\sin^2 3x + \cos^2 3x) + (\cos^2 3x - \sin^2 3x)}$

$= \dfrac{2 \sin 3x \cos 3x}{2 \cos^2 3x}$

$= \dfrac{\sin 3x}{\cos 3x} = \tan 3x \Rightarrow \theta = 3x$

$\Rightarrow r \operatorname{cis} 3x = 0 \Rightarrow r = 0$ of $\operatorname{cis} 3x = 0$

$\Rightarrow r = 0 \Rightarrow r = \sqrt{(1 + cos6x)^2 + sin^2 6x} = \sqrt{\cos^2 6x + \sin^2 6x + 1 + 2\cos 6x}$

$= \sqrt{2 + 2\cos 6x} = \sqrt{2}\sqrt{1 + \cos 6x}$

$\Rightarrow r = \sqrt{2}\sqrt{1 + \cos 6x} \Rightarrow \sqrt{2}\sqrt{1 + \cos 6x} = 0$

$\Rightarrow \sqrt{1 + \cos 6x} = 0 \Rightarrow 1 + \cos 6x = 0 \Rightarrow \cos 6x = -1$

$6x = \operatorname{Bgcos}(-1) \Rightarrow 6x = \pi + 2k\pi, k \in \mathbb{R} \Rightarrow x = \dfrac{\pi + 2k\pi}{6} = \dfrac{\pi}{6} + \dfrac{k\pi}{3}, k \in \mathbb{R}$

$$\Rightarrow \operatorname{cis} 3x = 0 \Rightarrow \cos 3x + i \sin 3x = 0$$

$$\Rightarrow \left\{ \begin{array}{rcl} \cos 3x & = & 0 \\ \sin 3x & = & 0 \end{array} \right. \Rightarrow \operatorname{cis} 3x \neq 0$$

References